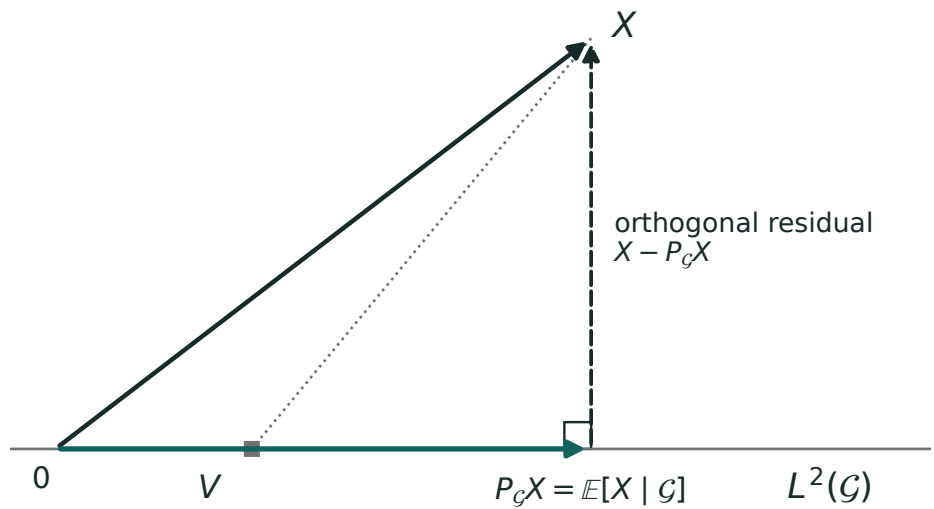
Conditional expectation as the best square prediction

$$\|X - V\|_2^2 = \|X - P_{\mathcal{G}}X\|_2^2 + \|P_{\mathcal{G}}X - V\|_2^2$$



Abstract geometry in a Hilbert space: the drawing is two-dimensional, not the space of all random variables.